

Appendix

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This Appendix contains additional data, sources, estimations and methodological details referenced to in the article *Was it really so high? New estimates on income inequality in Finland, 1865–1959*. The Appendix has been structured as follows. Section 1 describes the consent and municipal tax records in detail and goes through the generalised Pareto curve interpolation process step-by-step. Section 3 describes the subsistence based estimated for the article’s maximum feasible Gini coefficient. Section 4 presents additional data on the development of income taxation in Finland from 1938 to 1959. Section 5 presents the average equity ratio and rates of return on equity and total capital in Finnish industries in 1837–1938. Section ?? compares the top 1 percent’s income share to those of the bottom 50 percent and the top 25–2 percent, or the fractile called as the ‘middle class’ for short. The section includes additional description of the consent tax micro data. Finally, Section 6 provides a comparison of agricultural productivity in Finland and the U.S. in 1880.

1. Generalised Pareto curve interpolation

This appendix presents a detailed description of the methodology behind the income share estimates presented in the article, with a particular focus on consent and municipal tax records from 1865, 1871, 1876, 1880, and 1899–1900. Although these estimates are far from perfect, this appendix delves into some of the reasons why it may be reasonable to assume that the top 1 percent’s taxed income share was not at an historical apex during the latter half of the 19th century, but only ascendant.

The first nation-wide income tax (*suostuntavero, bevillning*) was levied in the Grand Duchy of Finland between 1865 and 1885. I have translated it very directly as ‘consent tax’. The name refers to the fact that it was levied on the estates’ consent, enabled by the convocation of the estates by tsar Alexander II in 1863. It should be noted that the term is not unique. It applies to a number of taxes levied in Finland both under Swedish and Russian rule. It was only in the 1920s that a more modern terminology of income taxation began to supersede different forms of ‘consent taxation’.¹ The tax was heavily influenced by a similar tax introduced in Sweden in 1861 (Nykänen, 2012, p. 55). It was somewhat progressive, with the statutory tax rates ranging from 0.8 percent for incomes above 500 FIM to 1.2 percent for incomes above 10,000 FIM. The tax rates and thresholds were retained throughout the period, which has subjected the taxpayers to at least some degree of ‘bracket creep’, that is, an increase in the tax rates in the absence of inflation checks.² This is visible in the growth in the number of tax units in the data, as their incomes have risen. The tax had to be renewed by the estates. Its abolition was thus a passive process once they declined to do so in 1885.³ A state income tax has been levied in Finland annually since its introduction in 1920.

¹The Diet (*säätyvaltiopäivät, Finlands ständer*) had been abolished already in 1906 and replaced by a single-chamber parliament. This speaks of the delay in the change of historically established terminology—after all, there was no longer any need for the ‘consent’ of the estates, which were *de facto* losing their status around the time.

²This phenomenon became much more prominent in the 20th century, in particularly during WWI and WWII in most warring countries, including Finland and Sweden (e.g. Henrekson and Stenkula, 2015, p. 13).

³According to Wikström (1985, pp. 29–30, 33), one issue with the tax was the relative evenness of income distribution which meant that the top tax rates did not bring in the anticipated revenue. In addition, although the vast majority of the Finnish population was rural, they accounted for less than a half of the revenue. Thus, the low yield was the primary reason for the abolishment of the tax.

The tax unit in the consent tax of 1865–85 was primarily the household head. However, as mentioned in the article, the tax concerned also corporations (*osake- ja pankki-yhtiöt*) with regards to their dividends in 1865–79 and their ‘pure’ income in 1880–5. In addition, estates of the deceased were also among the units. Although few in number, the article has shown that the corporations’ impact should not be neglected. Although the data analysed thus far covers only a small part of the taxed corporations, the estimations already show an impact on the top 1 percent’s share. The tax concerned only a relatively small number of households. In 1865, this figure was approximately 18.8 percent. In 1880, the figure had risen to 31.2 percent of the estimated number of households (Roikonen and Heikkinen, 2020, p. 49). The exclusion of the vast majority of households has boiled down on the relatively high income threshold of 500 FIM per annum.

The tax was levied by local tax committees with between five to nine members. Rural municipalities had a single committee, whereas towns and cities could have several based on their population. The committees were supposed to represent the different social groups in their district. The committees’ task was to assess the tax units’ ‘pure’ income based on available information, be it the taxpayers’ declarations or the quantity and value of production, sales opportunities, the scale and quality of enterprise, the value of parcels (i.e. wages in-kind), rates of interest, value of bequests, and so on (April 20, 1865/23, § 2). As will be shown later on, and as the article mentions, these information requirements made the task quite challenging.

Although taxpayers were obliged to file a tax return under the statute, there were no consequences for not doing so. Over time, the share of filers dropped quite significantly. This means that the tax committees assessed their incomes mainly on their own in the sense that they could not rely on such returns, for better or worse. For example, in 1866, when the tax was levied for the first time on the previous year’s incomes, around 26.0 percent of all units filed a return. In 1876, the same figure was merely 2.9 percent. It seems that, at least in 1865–6, more than a third, or 35.6 percent, declared incomes which were deemed too low by the local tax committees (9.3 percent of the total). It remains unclear to what degree this was motivated by tax evasion—after all, the tax ranged from merely 0.8 to 1.2 percent—or whether taxpayers contended with the committees’ assessments on pragmatic grounds (e.g. the low rates and high exemption threshold meaning that there were few incentives to spend time on compiling a return) (Statistical Office of Finland, 1869, pp. 9–10; Statistical Office of Finland, 1875, p. 10).

The concept of income in these early years was ‘pure’ income. According to the original statute for the years 1865–7,

Pure income is the saving which is created when some person’s fixed or movable capital, office or business, and the increase in her wealth has otherwise through inheritance, testamentary or interstate succession, estate, discovery, pensions, year of grace [in Finnish, *armovuosi*, in Swedish, *nåddår*] etc. received, are to be deducted not only annual expenses, wages, interests, pensions etc. but also all payments to the state or the municipality, irrespective of their name or quality. In addition, no deduction shall be made for the cost of living of the taxpayer and her family, the wage and maintenance of her various servants, or for any other charges relating thereto. Dividends received from companies which are subject to income tax under Article 3 [i.e. corporations] are, however, to be deducted when the shareholders’ personal incomes are calculated (March 2, 1865/10, § 4, the author’s translation).

As the statute did not include deductions, exemptions or imputations apart from the dividend deduction and a 500 FIM deduction for tax units whose income equalled or exceeded 550 FIM (the incomes were rounded to the nearest 100 FIM), the concept of ‘pure’ income was relatively close to more modern concepts of pre-tax or taxed income, including inheritances and capital gains, or profits from sales. Although the concept was modern, the income assessments, obviously, hinged on the accuracy of the local tax committees’ assessments. Some authors, including Roikonen and Heikkinen (2020) and Voutilainen (2022) have emphasised the accuracy of these assessments, given the fact that the committees comprised of individuals supposedly well versed in the tax subjects’ circumstances. This does not imply that there would not have been heterogeneity in the assessments. At this stage, it is hard to find evidence of any systematic biases in favour or to the detriment of certain tax units, although it is probable that the tax base was more complete for wages than, say, for rents (Statistical Office of Finland, 1875, p. 12). Regional differences may also have affected the assessments, and thus, effective tax rates to a degree.

The top income share and Gini coefficient estimates for the non-exempted part of the income distribution have been obtained using a so-called generalised Pareto curve interpolation developed by Blanchet et al. (2017). This method is known for its suit-

ability for the assessment of top incomes. By definition, a generalised Pareto curve comprises of inverted Pareto coefficients $b(p)$ following the rank $0 \leq p \leq 1$, where $b(p)$ is ratio between the average income above p and the p -th quantile $Q(p)$ so that $b(p) = E(X|X > Q(p))/Q(p)$. For example, in 1880, the average taxed income in the 10,000 to 50,000 FIM bracket was 19,462 FIM, so that the inverted Pareto coefficient between these quantiles, 10,000 and 50,000 FIM, was 1.9 at a single-decimal level. Blanchet et al. (2017) note the usefulness of this method for historical tax records because the interpolation requires only relatively little information in order to produce supposedly accurate estimates. They also laud the accuracy of the method, claiming that its precision ‘is such that it is often preferable to use tabulations based on exhaustive data sets rather than individual data from a non-exhaustive subsample of the population, even for subsamples considered very large by statistical standards.’ For example, in the case of a 100,000-observation subsample, the mean relative error for the top 5 percent’s share can be in the range of three percent, whereas the same for the tabulated data using the generalised Pareto curve interpolation can be as low as 0.5 percent with the difference increasing rapidly as one moves up the distribution. For the top 0.1 percent, the mean relative error can be as high as 20 percent in a 100,000-observation subsample, but as low as four percent in tabulated data using the given method (Blanchet et al., 2017, pp. 2, 3). Paretian interpolation methods have become the staple in historical income inequality research across the globe (e.g. Atkinson and Piketty, 2007; Atkinson and Piketty, 2010; Piketty and Saez, 2003). Although other tail-heavy distributions, such as the log-normal can also be used for similar interpolation purposes, the generalised Pareto curve interpolation seems to suit the purpose reasonably well.

The Statistical (Main) Office of Finland produced annual consent tax statistics for the taxed incomes in 1865, 1871, 1876, and 1880. The tabulations contain data on bracket thresholds (e.g. 10,000 to 50,000 FIM), the number of tax units within these thresholds (for example, 905 in the given bracket in 1865) by social group (among others, state officials, merchants, industrialists and peasants) by their primary occupation and their tax sums. As the income brackets follow the thresholds in the income tax statute (His Imperial Majesty’s Benignant Declaration on the consent tax that the Finnish Estates have taken upon themselves to pay in the years 1865, 1866 and 1867 (March 2, 1865/10)), it is possible to weight these rates with the number of tax units in each bracket in order to arrive at an estimate on the weighted average tax rate. The total sum of taxes can then be divided by this weighted average in order to obtain an estimate of the total sum of taxed income in the given year. Because this method captures only the taxed incomes above the 500 FIM exemption limit, each tax unit above the bottom threshold of 500 FIM have been assigned an additional 500 FIM in order arrive at the level of total ‘pure’ income.⁴ This can then be divided with the number of tax units (80,517 in 1865) in order to estimate the average ‘pure’ income (taxed income plus the 500 FIM deduction). More formally, this can be expressed as

$$\bar{y}_t = \left(\frac{\sum_{i=1}^N \tau_{it}}{\phi_t} + [500 \text{ FIM} \times \sum_{i=1}^N u_{it}] \right) / \sum_{i=1}^N u_{it}, \quad (1)$$

where $\bar{y}$ is the average ‘pure’ income, u is the tax unit, t is the year of observation, τ stands for the consent tax and ϕ is the weighted average tax rate using the number of tax units in each bracket as weights. For the year 1880, the data include sums of taxes paid by the tax units in each income bracket. This allows one to estimate the average taxed income slightly more reliably. Still, the differences are minor—in 1880, the estimated total taxed income is 162.2 million FIM when the bracket sums are aggregated, the 1880 data allowing for such a calculation. If one uses the weighted average tax rate of 1.0 percent as in Equation 1, the total taxed income is 156.9 million FIM. In per capita terms, the figures are 1,144 and 1,107 FIM, respectively, a 3.4-percent difference, which can be considered modest. Because there were no other exemptions than the one for incomes below 500 FIM and the ensuing 500 FIM exemption from the ‘pure’ income of tax units with incomes above this level, the total tax yield allows for a fairly reliable estimation of the taxed income, as well as the ‘pure’ income, when each tax unit are assigned an ‘extra’ 500 FIM.

⁴Dividing the total sum of taxes (e.g. 668,828 FIM in 1865) with the average weighted tax rate (1.0 percent from 1865 to 1880) yields only the sum of income above the 500 FIM exemption threshold. For example, if a household head, i.e. the intended unit of taxation alongside corporations, had an assessed income of 700 FIM, the first 500 FIM would have been exempted and the tax levied only on the remaining 200 FIM. The tax rate below 5,000 FIM being 0.8 percent, the tax would have in this case been 1.6 FIM (0.008×200 FIM). In total, the taxed income above the 500 FIM threshold would have been around 82.9 million FIM in 1865. When each tax unit has been assigned an ‘extra’ 500 FIM, the total taxed income, or the ‘pure’ income is 123.1 million FIM.

In practice, the share estimates presented in this article have been produced using the `gpinter` package written by Thomas Blanchet for R. The function `thresholds_fit` has been used in particular. This function requires data on percentiles, bracket thresholds, the average income within the distribution (obtained using the method described previously in this appendix), and, preferably, data on the bottom threshold (in this case, 500 FIM) and the top bracket average or income share (100,000 FIM or above). As this data cannot be obtained straight away for 1865, 1871, and 1876, it has been estimated using the quotient of the top bracket average to the total distribution average in 1880 (i.e. 0.005) by dividing the average income (see column four in Table 1 below) with this figure.

Table 1 shows the data needed for the generalised Pareto curve interpolation for the year 1865.

Thresholds (FIM)	Tax units	Percentile	Tax rate (%)	Tax yield (FIM)	'Pure' income (FIM)
500	1,000	60,941	75.69	0.80	
1,000	5,000	17,512	97.44	0.80	
5,000	10,000	1,369	99.14	1.00	
10,000	50,000	662	99.96	1.20	
50,000	100,000	28	99.99	1.20	
100,000	5	100.00	1.20		
	80,517		1.00	8	1,530

Table 1: An example of tabulated tax data in 1865. Source: Statistical Office of Finland, 1869, Table 1, pp. L–LI. Note: the bottom row of the fifth column gives the weighted average tax rate using the number of tax units in each bracket as weights. The bottom rows of the sixth column gives the average tax and 'pure' income per capita. These data include corporations and their dividends. Note also that the data have not been adjusted for the tax units below the 500 FIM threshold as, for example, in Roikonen (2022), where an additional bracket is added below the 500–1,000 FIM bracket and assigned an average income of 40, 60 or 80 percent of the 500 FIM threshold.

The total tax yield in 1865 was 668,828 FIM. Using the 1.0 percent weighted tax rate average, this produces a total income of 82,893,448 FIM as these figures' quotient. Given the 500 FIM exemption, adding this amount of income to the taxed income of each unit, the 'pure' income is 123,151,948 FIM, or by 48.6 higher than the sum of taxed income. Given the number of tax units (80,517 in the given year), this results in an average income of 1,530 FIM per tax unit, which is 500 FIM or 48.5 percent more than the average for taxed income, 1,030 FIM.⁵ Thus, moving from the taxed income to 'pure' income increases the total and per capita incomes quite significantly. Not only this, but it also results in lower inequality, as the addition of the deducted 500 FIM to each tax units' income is relatively a greater increase at the bottom than at the top of the distribution.⁶ The ratio of the total average income to the average in the top bracket in 1880 was 228,584 FIM to 1,144 FIM, or 0.005. This figure has been used to divide the 1,530 FIM average in 1865, thus obtaining a top bracket average of 305,903 FIM.

The corporations' dividends and, in 1880, their 'pure' income, has been subtracted by assigning each recorded corporation and its dividend to its 'respective' income bracket, after which their dividends have been summed up and they have been subtracted from the number of tax units per bracket and their sum of dividends has been subtracted from the sum of 'pure' income. Given the available data from the financial statements, the number of tax units decreases to 80,513. The total 'pure' income declines to 122,380,848 FIM, or by 0.6 percent. The average 'pure' income per capita declines to 1,520 FIM. Although this may not seem much, the relevant changes happen at the very top, or within the top income bracket. Note that there were five tax units in this bracket in 1865. Given the data on corporations' dividends available at this stage, at least three of should have been corporations—if the local tax committees have identified them and taxed them accordingly. Their total dividend was 768,000 FIM. When this sum is removed from the total 'pure' income in the bracket, the average income actually increases to 379,207 FIM. This is because

⁵Note that thresholds have been adjusted upwards accordingly. For example, the 1,000 FIM threshold has become 1,500 FIM, and so on. This is necessary for the generalised Pareto curve interpolation to function properly, as the average income of 1,530 FIM would not otherwise match these bracket thresholds.

⁶This can be illustrated accordingly. If we take a vector of seven incomes (100, 200, 500, 1000, 2000, 5000, and 10,000 FIM), their Gini coefficient is 0.620. If we add 500 FIM to each of these incomes, the Gini coefficient is 0.523, or 0.097 points lower than the initial value.

dividing the remaining income with the other two tax units results in a higher average. Nevertheless, subtracting the corporations brings about a lower level of inequality, as there are fewer tax units and less income at the top of the distribution.

Subtracting the corporations and their incomes from the tabulations becomes much more straightforward in the case of the 1899–1900 municipal tax data (A. Hjelt and Broms, 1904, 1905), as the tabulations themselves indicate their numbers, incomes and taxes. When the new income and wealth tax was introduced in 1920, the corporations were reported separately, which homogenises the data even further in terms of tax units, although the unit itself would continue to change between individuals and jointly taxed married or jointly self-employed couples (Jäntti et al., 2010, Table 8A.9, p. 432). These issues aside, most of the estimation issues discussed in the article as well as in this appendix concern the initial four years of this study.

As has been mentioned in the article and in this Appendix, dividends were not subjected to double taxation under the consent tax. Instead, they were taxed at the level of the corporation. In 1865–79, corporations were taxed only for their distributed dividends. From 1880 to 1885, corporations, as other tax units, were taxed for their ‘pure’ income. This makes their subtraction from the tax unit population all the more important. If one chooses to follow the concept of ‘pure’ income at the level of household heads—i.e. the ‘ideal’ homogeneous tax unit population in the sense that the tax was, in principle, and alongside the aforementioned corporations, supposed to be levied on their income—then the corporations’ dividends and ‘pure’ incomes are supposed to vanish in thin air. However, if one wishes to broaden the concept and see how the dividends (and, at this stage, only dividends, for the undistributed profits have not directly formed a part of the shareholders’ personal incomes, although using a broader concept of income, e.g. that in contemporary distributed national account, or DINA, studies, (for example, Garbinti et al., 2018) would see these undistributed profits measured as part of their income) have affected the distribution of income at household level, one needs to estimate their distribution.

As I discuss in the article, the corporations’ dividends and ‘pure’ have been subtracted from the data in an indirect manner by means of their financial statements. As such, the study captures only a part of the said incomes. Micro data, that is, the consent tax accounts, are of little use here, because they only show the tax units’ total income depending on the tax committee. In other words, the accounts do not break the incomes down to separate items, such as wages and salaries, entrepreneurial and capital income. The situation is no better in the case of the municipal tax data in 1899–1900, although in this case the corporations’ incomes can be subtracted from the tabulated data directly, which means, in principle, that their subtraction offers a truer picture of income inequality among household heads. An issue in this latter case, however, is that the data does not break the corporations’ ‘pure’ income down to dividends and undistributed profits (e.g. deposits to profit, pension, insurance, and auxiliary accounts). I have tried to estimate the share of dividends in their net profits with the financial statements data I have for industrial corporations and banks. I have calculated a weighted average pay-out share, which in the case of a the i -th corporation translates to

$$\kappa'_{it} = \left(\frac{d_{it}}{(d_{it} + \theta_{it})} \right) \left(\frac{d_{it}}{\sum_{i=1}^n d_{it}} \right), \quad (2)$$

where $i = 1, \dots, n$ are the corporations, d is the dividend and θ is the undistributed profit, and the factor in the second round brackets is the weight of this given corporation. In 1899, this weighted average pay-out share was 0.748 and in 1900 it was 0.840. Thus, I have estimated that 74.8 percent of the corporations ‘pure’ income was paid out in dividends in 1899 and 84.0 percent in 1900. One issue with this method are outliers: extremely high pay-out shares due to losses. In 1899, the data comprises of 13 corporations, five of which turned a loss. In 1900, six out of the 14 corporations did the same. In the case where the said corporations both turned a loss and paid out an even more substantial dividend—an example of this being Finlayson in 1900—the pay-out share can appear extremely high (in this case, 41.250). I have omitted such cases when estimating the weighted average, which lowers its level somewhat.

As mentioned in the previous paragraphs, neither the published consent tax tabulations nor the micro data tell us anything about the distribution of different income items. Therefore, one needs to rely on external sources. I have estimated the distribution of

the dividends among household heads by using the closest available data, which in this case means the tabulations on complete tax returns ($n = 470,431$) in the 1921 income and wealth tax statistics (OSF IV B:1, Table 6, pp. 30–31). This data is admittedly quite distant from the turn of the century, not to mention 1865. Still, it is the nearest disaggregated data available. I proxy the distribution of dividends using that of capital income (rents, interest and dividends). Fortunately for the municipal tax records in 1899 and 1900, this 1921 data distinguishes between rural and urban municipalities. I have therefore been able to produce separate estimations for each sub-group of data. I have then used the share of tax units in rural and urban municipalities as weights for arriving at a weighted average income share of the top 1 percent, as well as the total income and average.

From here on, the method is simple. I assume that the distribution of dividends in 1865–1900 has been identical to that of capital income in 1921 and allocate the corporations' dividends accordingly to the top 1 percent. In 1921, the top 1 percent's capital income share was 23.8 percent. I thus assume that this share of the dividends I have managed to find in the archives has gone to the said income group. For example, in 1865, 183,525 of the total of 771,100 FIM has been allocated to the top 1 percent. This method is, of course, rather coarse. For example, it does not account for dividends between non-household owners or foreign owners. Hence, even though I have only managed to obtain an unknown share of the dividends (at least those of the biggest industrial corporations and two major banks), the method presumably exaggerates the share of dividends received by the top 1 percent of household heads. Furthermore, because the 1921 data does not break down the capital income into its sub-components, it is difficult to say if dividends were more or less unevenly distributed than capital income in total.

A critical question with the 1921 data is timing. Did WWI, and, in the Finnish case, the civil war of 1918, alongside with the tumultuous aftermath of these epochal events, still impact the level and distribution of income three years later? In other words, can these data be reasonably compared to the observations from 1865, 1871, 1876, 1880, and 1899–1900? According to Heikkinen (2017, p. 68), the Finnish exports were by 1921 following a relatively strong path to recovery. In addition, prices, profits and the GDP did not contract in the newly-independent Finland after the war—on the contrary, the wholesale price index increased almost twelvefold in 1914–21, while the GDP, in nominal terms, grew from 1,192 million FIM in 1914 to 12,095 million in 1921 (Vattula, 1983, Table 10.2.a, pp. 375–377 and Table 14.5.b, p. 459). At the same time, the studied corporations' nominal net profits grew from 14.5 million in 1914 to 125.8 million in 1921 (BOF Teollisuuden; NHA Suomen). In relative terms, however, these corporations' profits-to-GDP ratio actually fell from 1.2 to 1.0 percent. In 1865–1900, based on the six years available, the ratio was 0.8 percent. Therefore, although the ratio was higher in 1921 than in, say, 1865, when it was a more modest 0.4 percent (probably the result of low incorporation rate), it seems to be so that the level of profits in 1921 was no longer affected either by the war-time peak demand (the ratio increased from 1.2 percent in 1914 to 2.7 percent in 1916) nor its collapse to 0.5–0.6 percent in 1917–8. Hence, the 1921 data can be seen as 'normal' in the sense that the profits-to-GDP ratio was closer to its level at the start of the war.

2. A comparison with micro data

How well do these interpolation-based share and inequality estimates match with the underlying micro data? Table 2 compares them with individual tax assessments from the consent tax accounts in 1876. These include the taxpayer's name, their title, total income assessed by the tax committee, their total income in the tax return (if any), and occasional notes by a supervisory committee. Finally, the accounts give tax to be paid by the taxpayer. The reason for choosing this year over the others is the onset of major-scale industrialisation around the time (Heikkinen, 1986, p. 129), as well as the fact that the country had by then recovered from the 1867–8 famine. They should thus allow to assess the impact of corporation on inequality under relatively 'normal' circumstances. The accounts are the basis for the tabulations which have been used both in Roikonen and Heikkinen (2020), Roikonen (2022) and in this study. The data come from the rural municipalities Iisalmi and Sipoo, located in eastern and southern Finland, respectively, and the industrial town of Tampere north of Helsinki. Iisalmi is an example of a rural locality in one of the poorest regions of Finland, Savonia. Sipoo, on the other hand, was a relatively prosperous coastal community east of the capital Helsinki. Tampere is a good example of the impact of corporations on income inequality because many of Finland's largest industrial corporations were located there, including Finlayson and the paper manufacturer J. C. Frenckell & Son. As the micro data lack tax units below the 500 FIM threshold, the figures have not been adjusted accordingly (see the Appendix for further

discussion). This applies also to the Pareto interpolation, which estimates the figures only for the tax units above the exemption threshold (which, in practical terms, is 600 FIM, as the tax committees rounded the incomes above 550 FIM to the nearest one hundred FIM). Thus, the exempted units are not accounted for in either case.

Item	Interpolated	Interpolated (w/o corp.)	Micro data	Micro data (w/o corp.)
<i>Iisalmi (n = 695)</i>				
Top 1% share	6.2%	n/a	8.4%	n/a
Top 1% average	6,070	n/a	11,000	n/a
Total average	980	n/a	949	n/a
Tax yield	2,678	n/a	2,681	n/a
Total income	681,170	n/a	659,700	n/a
Gini	0.155	n/a	0.295	n/a
<i>Sipoo (n = 364)</i>				
Top 1% share	7.9%	n/a	8.0%	n/a
Top 1% average	8,211	n/a	9,333	n/a
Total average	1,041	n/a	963	n/a
Tax yield	1,515	n/a	1,493	n/a
Total income	363,309	n/a	350,600	n/a
Gini	0.132	n/a	0.290	n/a
<i>Tampere (n = 873 & 864)^(a)</i>				
Top 1% share	49.9%	39.0%	38.8%	18.1%
Top 1% average	143,497	59,587	105,500	33,000
Total average	2,877	1,530	2,494	1,692
Tax yield	17,201	15,893	18,379	10,229
Total income	2,549,148	1,831,248	2,176,900	1,461,900
Gini	0.609	0.522	0.694	0.558

Table 2: Robustness checks for the interpolations using data from Iisalmi, Sipoo and Tampere, 1876 (FIM). Source: Statistical Office of Finland, 1880, National Archives (NA), Kuopion läänintilit (Ga:310), Suostuntaverojen tilit 1877 (I Gac:13), Suostuntaverotilitykset 1877 (Glc 17). Note: (a) with and without corporations. The concept of income is taxable income. As there are tax units in only the first four brackets in Iisalmi, Sipoo and Tampere (w/o corp.), a ratio of the total average income to that in the fourth bracket in the 1880 consent tax tabulation (0.058 to 1) has been used to estimate the average income in this (now highest) income bracket. ‘Interpolated’ refers to the generalised Pareto curve interpolation.

Overall, income inequality seems to have been lower in Iisalmi and Sipoo based on the interpolation than on the micro data. In Iisalmi, the top 1 percent’s share is 2.2 percentage points lower than in the micro data, with the corresponding average being 4,930

FIM lower. The total or per capita average, however, is much closer, which suggests that dividing the tax yield by the weighted average tax rate allows one to make reasonable estimations of both this average and the total taxable income, the latter being only around 3.2 lower in the micro data. The Gini coefficient, on the other hand, is 140 Gini points lower based on the generalised Pareto interpolation than in the micro data. Together, both the low share of the top 1 percent and the Gini coefficient make Iisalmi appear much more equal based on the Pareto interpolation than is the case with the micro data. In the case of Sipoo, the top 1 percent's share is almost identical, but the average income within this bracket as well as the Gini coefficient are much lower, although the total average is only around 7.5 percent higher based on the Pareto interpolation, with the estimated total taxable income being approximately 3.5 percent higher based on the interpolation. The situation is opposite in Tampere, where the results are mixed. The top 1 percent's income share is 11.1 percentage points higher based on the interpolation on the whole tax unit population. When corporations are removed from the data, the top 1 percent's share falls by 10.9 percentage points, but not as much as in the micro data, where the top 1 percent's share is 20.7 percentage points lower once the corporations have been removed. The average incomes within the top 1 percent, on the other hand, are much close to one another, as are the total averages, tax yields, and total incomes. The Gini coefficients, too, are closer to one another, with the coefficient for the entire tax unit population, including corporations, being only around 85 Gini points or 14.0 percent higher in the micro data, and only 36 Gini points (6.9 percent) higher once the corporations have been removed.

These comparisons show that the inequality estimates based on the generalised Pareto curve interpolation are sensitive to the data. In the tabulated data for Tampere, there are tax units in each income bracket, which probably improves the reliability of the estimates relative to Iisalmi and Sipoo. To conclude, this method appears to exaggerate the top 1 percent's income share and average in Tampere, whereas the opposite is true in Iisalmi and Sipoo, which also appear very equal in terms of the Gini coefficients from the Pareto interpolation.

These results do not imply that the micro data are unproblematic themselves. On the contrary, they appear worryingly homogeneous, with a relative majority of the tax units having been given exactly the same incomes by the tax committees (e.g. 600, 700, or 800 FIM), as seen on Table 3. The tax committees, it seems, have made very similar assessments for a great share of the population. One reason for this is that they had to round their assessments to the nearest hundred FIM. Still, a bigger reason may be their lack of knowledge on the taxpayers' incomes. For example, 38.1 percent of the tax units in the micro data have an income of 600 FIM, 22.6 percent have an income of 700 FIM, and 10.6 percent an income of 800 FIM. It cannot be said at this stage whether his homogeneity stems from any systematic biases in the data, such as persistent under- or over-assessments by the committees. It is also difficult to say whether the Finnish independence in 1917 has influenced the collection of taxes and, subsequently, the income assessments of the local tax committees from 1920 to 1959. One can therefore wonder if the interpolation is capable of estimating the 'true' income shares and inequality actually better than the micro data, although this remains speculative at this stage.

Income (FIM)	<i>n</i>	Share
600	736	38.1%
700	437	22.6%
800	205	10.6%
900–1,500	263	13.6%
1,500–2,000	84	4.4%
2,000–5,000	129	6.7%
5,000–10,000	42	2.2%
10,000–50,000	29	1.5%
50,000–100,000	4	0.2%
>100,000	1	0.005%
total	1,930	

Table 3: The distribution of tax units within income brackets (FIM), including corporations, in Iisalmi, Sipoo and Tampere, 1876. Source: NA Ga:310, I Gac:13, Glc 17.

In the article, the estimated income shares, averages, Gini coefficients and other figures were compared to those obtained from the consent tax accounts of 1876, using data from Iisalmi, Sipoo and Tampere. Table 4 complements those comparisons with additional shares and averages. These include, alongside the already presented figures for the top 1 percent, the ‘middle class’, i.e. the tax units between the 60th and 90th percentile and the bottom 50 percent. As before, the ‘n/a’s’ refer to the fact that there have been no taxed corporations in Iisalmi and Sipoo in 1876, while in Tampere there was nine.

Item	Tabulated	Tabulated (w/o corp.)	Micro data	Micro data (w/o corp.)
<i>Iisalmi (n = 695)</i>				
Top 1% share	6.2%	n/a	8.4%	n/a
Top 60%–90% share	30.7%	n/a	28.6%	n/a
Bottom 50% share	40.5%	n/a	43.3%	n/a
Top 1% average	6,070	n/a	11,000	n/a
Top 60%–90% average	1,230	n/a	1,247	n/a
Bottom 50% average	980	n/a	600	n/a
Total average	980	n/a	949	n/a
<i>Sipoo (n = 364)</i>				
Top 1% share	7.9%	n/a	8.0%	n/a
Top 60%–90% share	28.3%	n/a	19.4%	n/a
Bottom 50% share	42.6%	n/a	38.5%	n/a
Top 1% average	8,211	n/a	9,333	n/a
Top 60%–90% average	1,255	n/a	1,079	n/a
Bottom 50% average	1,041	n/a	600	n/a
Total average	1,041	n/a	963	n/a
<i>Tampere (n = 873 & 864)^(a)</i>				
Top 1% share	49.9%	39.0%	38.8%	18.1%
Top 60%–90% share	15.4%	19.4%	15.9%	20.4%
Bottom 50% share	18.1%	21.9%	9.2%	13.7%
Top 1% average	143,497	115,316	105,500	33,000
Top 60%–90% average	1,476	987	1,628	1,499
Bottom 50% average	1,043	670	600	600
Total average	2,877	1,530	2,494	1,692

Table 4: Additional income share and average data, Iisalmi, Sipoo and Tampere, 1876 (FIM). Source: Statistical Office of Finland, 1880, National Archives (NA), Kuopion läänintilit (Ga:310), Suostuntaverojen tilit 1877 (I Gac:13), Suostuntaverotilitykset 1877 (Glc 17). Note: (a) with and without corporations.

In Iisalmi, the bracket shares appear lower for the top 1 percent and the bottom 50 percent based on the interpolation than in the micro data. The opposite holds true for the bracket average incomes, apart from the total per capita average, which is 31 FIM lower in the micro data. In the case of Sipoo, the difference in the top 1 percent’s share is only 0.1 percentage points, whereas the share

of the ‘middle class’ is 8.9 percentage points below that in the micro data. The bottom 50 percent’s share, on the other hand, is 4.1 percentage points higher based on the interpolation. In the case of Tampere, the situation is again quite different. Here, the top 1 percent’s share appears much higher based on the interpolation than in the micro data. In the micro data, without corporations, the share of the ‘middle class’ grows by 4.5 percentage points. The share of this bracket does not respond as strongly to the way the corporations have been removed from the tabulated data. These differences suggest that the article’s method may not adjust the top income shares sufficiently, even if all corporations could be accounted for, as in this case, where the corporations in the micro data have been joined to the tabulation for better comparability. An interesting peculiarity in the case of Tampere is the fact that the removal of the corporations actually seems to *increase* inequality instead of decreasing it. This cannot obviously be generalised upon, but it shows that the their removal may also have an opposite effect on estimated income inequality, as well.

The income share and inequality estimates presented in this study can, at best, be only as good as the data allows. For example, one may wonder whether 60.7 percent of the tax units really had an income which rounded up to 600 or 700 FIM, respectively. While we may appreciate the almost insurmountable difficulties that the local tax committees have faced when assessing people’s and corporations’ incomes, it may well be that the ‘true’ level of inequality was, in 1876 as in the other years during the late 19th and early 20th centuries, at least somewhat higher than the micro data suggests. Whether the generalised Pareto curve interpolation manages to produce estimates which are closer to this ‘true’ level of inequality remains an open question. I may well be that the consent tax records, at least, need to be supplemented with additional data not only for the income units below the 500 FIM threshold—i.e. the majority of the population between 1865 and 1885—but also for the units in the micro data itself.

3. Subsistence basket for the maximum feasible Gini coefficient

The level and trend of the maximum feasible Gini coefficient developed by Milanovic, Lindert, and Williamson (e.g. Milanovic, 2009, 2013; Milanovic et al., 2011) depends greatly on the level of subsistence. Although this level can be defined mechanically—for example, by assuming that the level is a certain percentage of the median income in a society—it can also follow the value of a so-called subsistence basket (for example, Allen and Weisdorf, 2011). This basket should, ideally, comprise of goods which have been deemed socially and physiologically necessary for the reproduction or survival of individuals and households in a given period of time. These goods should also be in realistic quantities and prices. One can also make a separation between a ‘bare-bones’ and a ‘respectability’ basket, the latter containing goods and quantities deemed fitting for a somewhat better off person or household. Although the literature and discussion on these kinds of consumption baskets is rich, to my knowledge there are no recent studies on the topic in Finland. I have thus chosen to rely on the classic, albeit problematic study by Hjelt (1912), which offers the nearest relevant point of departure for constructing such a basket. Hjelt studies industrial workers in Finland during the early 1900s. The study comprises of household accounts kept by worker households which were deemed ‘normal’ or typical (V. Hjelt, 1912, p. 22). The averages drawn from this accounts, therefore, cannot be deemed representative of the consumption of the population as a whole, but they do presumably capture as least many typical consumption goods and their quantitative relations to one another (e.g. bread butter, meat, and coffee). However, one needs to be quite careful when it comes to using these quantities, which the study acknowledges (V. Hjelt, 1912, pp. 76–77). Upon estimating the energy content (in kcal) of the items in an aggregate table of foodstuffs consumed by a sample ($n = 210$) of ‘normal families’ (V. Hjelt, 1912, p. 79) using modern data (U.S. Department of Agriculture, 2024), it became obvious that they could not have supported such a family, which, on average, comprises of 5.3 individuals or 2.6 adult male equivalent consumption units. Using more modern energy requirement figures (World Health Organization, 1985) for adult men and women engaged in manual agricultural labour as a reference point, it seems that the number and quantity of foodstuffs in the said table could only provide around 56.0 and 69.7 percent of the said men’s and women’s energy requirements per year, or 1,558 and 1,252 kcal per day versus the 2,780 and 2,235 kcal mentioned in the World Health Organization’s study.⁷ I have thus multiplied by the quantities in the aforementioned table by the ratio of these referential energy requirements to the ones obtained from the table, 1.78, to arrive at what can be deemed a subsistence level of food consumption. This increases the average per capita energy content of the basket from 433,521 to 773,709 kcal a year.

⁷Lundh (2013) uses this same study for comparing the standard of living between urban and rural households in southern Sweden in 1914 and 1920.

My aim has been to construct a basket for the benchmark year 1875. The reason for this is partially related to the availability of price, harvest, and trade data, although, in part, it has been a matter of choice. The basket has been constructed using data from the province of Uusimaa (in Swedish, Nyland), primarily due to **tehe availability of data**. I have adjusted the basket for the consumption of a rural household. This has necessitated a number of changes to the goods consumed by the 210 ‘normal’ households in Hjelt (1912). For example, instead of more processed goods, such as bread, cheese, sausages, and cold cuts, I have used grain and raw meat, for their processing has more likely taken place in the household (e.g. grinding), or the produce has been mainly sold instead of being consumed at the household (for example, cheese and butter). In addition, I have omitted coffee from the basket, because its consumption became more widespread only from the 1880s and 1890s onwards (Heikkinen, 1986, p. 127). Furthermore, it cannot be considered as a subsistence item relative to many other goods in the basket. I have, however, included sugar, for it was already imported and consumed around the time in reasonable quantities (Statistical Office of Finland, 1878, Table 18, p. 21). Whether it can be considered as a subsistence good at the time, is more debatable, although it was quickly being established as a common consumption item (Ikonen, 1991, pp. 32–33). Ideally, one would change the composition of the subsistence basket over time following changes in consumption and the standard of living. Indeed, this can be considered as a necessary step in future research. At this stage, and for the simple purpose of giving the maximum feasible Gini coefficient and the inequality extraction ratio a more solid footing, I have simply extrapolated the per capita price of the basket backward and forward using the GDP price index (1926=100) in the historical national accounts (Statistics Finland, 2024).

Good	Unit	Amount	Price (FIM)	Kcal
Grain ^(a)	kg	41.98	14.80	252,876
Milk and cream ^(b)	litre	104.06	59.25	153,958
Butter	kg	10.22	13.17	170,459
Eggs	dozen	0.75	0.94	770
Meat and lard ^(c)	kg	15.85	9.42	96,441
Fish ^(d)	kg	2.55	0.76	9,364
Potatoes	kg	33.65	4.60	60,258
Peas	kg	0.64	0.66	1,207
Sugar	kg	3.04	16.05	28,376
Rent ^(e)		4.6%		
Firewood	fathom	2.25	1.66	
Salt	litre	0.25	0.19	
Tallow ^(f)	kg	2.45	1.81	
Total			128.94	773,709

Table 5: Per capita annual subsistence consumption basket, 1875. Source: Allen and Weisdorf, 2011; Björkqvist, 1958; V. Hjelt, 1912; Lilius, 1888; Rabb, 1975; Statistical Office of Finland, 1878; Tilastollinen toimisto, 1879; U.S. Department of Agriculture, 2024; World Health Organization, 1985; National Archives (NA), Kamarikonttorin arkisto. E SAAPUNEET ASIAKIRJAT Eq Verohinnastot Eq:23; *Uusi Suometar* (January 24, 1879). Note: using price data from the province of Uusimaa in 1875; (a) the weighted average of the price and kcal content of barley, corn, rye and wheat based on their percentage shares of the arable in Sipoo, 1875, and their tax prices in the same year; (b) assuming that 95 per cent of the amount has been milk and five per cent cream; (c) assuming that 50 per cent has been meat (beef) and 50 per cent fat; (d) salted herring; (e) imputed per capita share of rent based on the household size in Hjelt (1912); (f) Allen and Weisdorf (2011, Table 1, p. 718) assume a consumption of 2.6 kg of candles per basket.

Unlike the earlier study by Parikka (1994, p. 57), who uses a so-called Hjelt’s minimum—an example of a single married couple living primarily on the husband’s income from masonry in Helsinki, 1900–8—the aim in this study has been to construct a subsistence consumption basket which captures more realistically the items, energy requirements, and cost of such a basket among the predominantly rural population in Finland around 1875. Whether this basket succeeds in this purpose is beyond the scope of this study. At the very least it can be claimed, however, that basing the basket on a larger, even though unrepresentative sample of

‘normal’ households, is a step forward from a somewhat arbitrary choice of a single example, as is the case with the aforementioned study by Parikka.

An obvious challenge with this basket is the fact that Finland was going through profound socio-economic changes during the late 19th and early 20th century. Relative, yet by no means complete self-sufficiency in the countryside was rapidly giving way to dependence on commodities bought in the market. Domestic produce was being replaced by industrial commodities. Items which had previously been considered luxuries (e.g. coffee, pastries, and soft beverages) were becoming normalised, even though their share in the households’ consumption remained diminutive. One particular problem with this and many other baskets is the omission of clothing and footwear, which were not only essential against cold and harsh weather, but also from a social point of view, as a signal of wellbeing and status. In future research, these and other changes in the composition of the basket should be accounted for.

4. Tax rates

The Gini coefficient(s) estimated for this study show a relative stable level of inequality in Finland **apart from the period from the mid-1930s to the 1940s**. This result seems to contradict recent views on WWI and WWII, in particular, as ‘great levellers’ due to war-time destruction of capital, loss of life and, especially, the increase in tax rates (e.g. Scheidel, 2017, p. 153). This does not mean that the main driving cause according to this explanation, taxation, had not increased in Finland during and after WWII. Indeed, as Figure 1 shows, both the top statutory marginal tax rate (MTR) and the effective tax rates in the top 0.1 and top 1 percent, the ‘middle class’, that is, the top 25–2 percent, and the bottom 50 percent did increase, albeit unevenly, from 1938 to 1948. Likewise, although state income taxation seems to have become less progressive from 1938 to 1945 according to the Kakwani index⁸ (admittedly, there are only three observations), the increase in tax rates in 1946 seem to have made more progressive in turn.⁹ The gist here is, however, that the changes in the effective tax rates and the progressivity of the tax system were relatively modest until the very end of the 1950s, when we observe a significant increase in the former and a decline in the latter. At the same time, the top MTR was cut in 1947, 1951, and 1953, thus pushing it closer to its **wartime** level.

⁸The Kakwani index measures tax progressivity as the difference of the concentration index of taxes and the Gini coefficient of pre-tax income, $P = C - G$. The higher the level, the more progressive the tax system. If P equals zero, the tax system is proportional and when it receives negative values, the system is regressive (Kakwani, 1977, p. 73).

⁹In addition to the state income tax, individuals, jointly taxed couples and corporations have paid a municipal income tax, which adds to these rates. However, because this tax has been proportional with some deductions for low-income taxpayers, its inclusion would probably lower the level of progressivity.

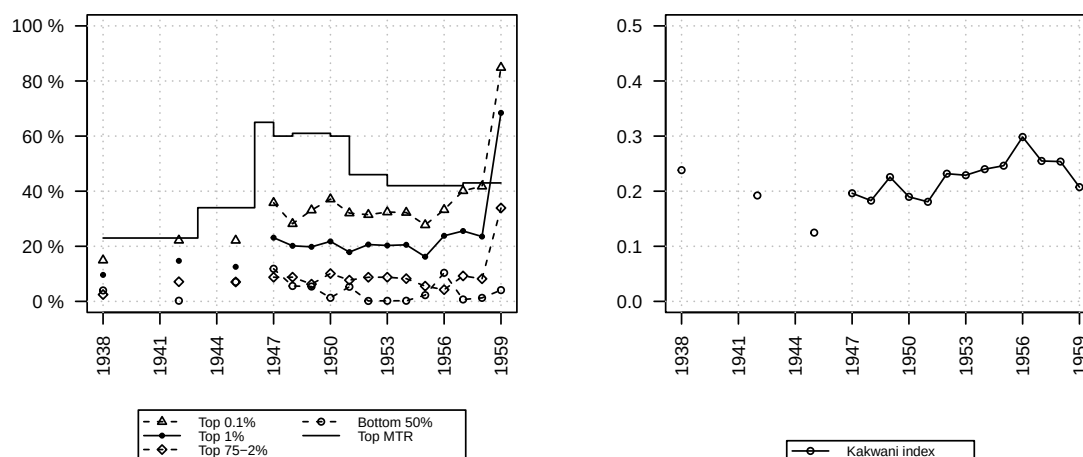


Figure 1: Effective tax rates the top statutory marginal tax rate and the Kakwani index, 1938–59. Source: Statistical Main Office of Finland, 1926–1963; AsK 1935 III–AsK 1958 III.

Although the changes the Gini coefficients or income inequality as a whole cannot be reduced to changes in taxation, the fact that the effective tax rates remained relatively stable between 1947 and the beginning of the 1950s may have been a contributing factor to the ‘uneventful’ development of the said coefficients, in addition to the opposite changes in the shares of the top 1 and bottom 50 percent, on the one hand, and the ‘middle class’ or the top 25–2 percent, on the other.

5. Equity ratio and rates of return in Finnish industries, 1837–1938

A possible explanatory variable for top income shares, such as that of the top 1 percent, is the rate of return on capital. Although it is difficult at this stage to say much about this or these rates, an indirect piece of evidence of it is the equity ratio, or the ratio of equity to total capital. Figure 2 presents this ratio for the industrial corporations used in this study in 1837–1938. The figure shows quite clearly, that the ratio remained quite stable with short-term fluctuations until the very end of the 19th century. The ratio declined from 1898 to 1917. After the Civil War of 1918 and a tumultuous adjustment period, the ratio recovered somewhat before declining in the 1930s. This implies that the corporations were increasingly unable to meet their investment requirements by means of undistributed profits and were becoming gradually more indebted. Whether this had repercussions on dividends and top incomes is uncertain at this stage. The rates of return on equity and total capital (equity and debt) remained relatively stationary with only short-term fluctuations from c. 1861 to 1908. In the 1910s, the rates appear, on average, higher than during the previous decades. There are two prominent peaks in these series in 1915–6 and 1919–20. These coincide, first, with the war-time

demand during WWI and the period of high inflation after the war.

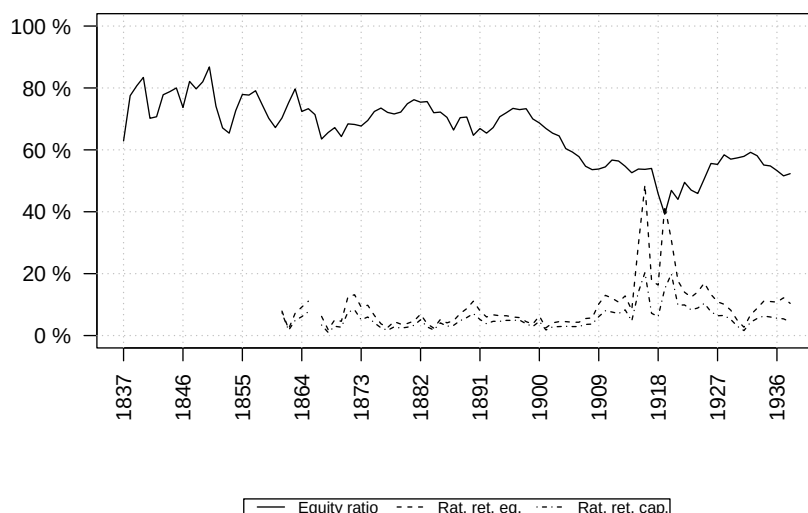


Figure 2: Weighted average equity ratio in Finnish industries, 1837–1939. Source: BOF Teollisuuden. Note: the figure uses the corporations' capital shares as weights.

6. Agricultural productivity in Finland and the U.S., 1880

Table 6 shows a comparison of agricultural productivity in Finland and the United States in 1880. The amount of arable and the total harvest per capita seem to have been around seven and five times higher in the U.S. Still, it is worth noting that the differences in grain yields are not as significant, apart from potatoes, where the difference is around fourfold. In the case of rye and wheat, the Finnish harvest figures actually show a higher yield than in the U.S. To what degree these differences are due to annual fluctuations has not been studied at this stage. Nevertheless, the table shows clearly that the average U.S. person occupied in agriculture worked on around seven times as much arable as in Finland at the time. The small scale of Finnish agriculture was one of the prime causes for its low productivity. Obviously, there are several complicating factors bringing uncertainty to these kinds of comparisons. First, the U.S. at the time, as it does now, spans an entire continent with multiple climate areas, significant differences in precipitation, humidity and temperature, soil quality, and so on. The U.S. as a whole, apart from Alaska, is located at a more southerly latitude than Finland. Secondly, the statistics themselves add another layer of complexity. In the Finnish case, for example, the cultivation and harvest data in the *Statistical Yearbooks* give only the most prominent crops, not all cultivated or harvested crops as a whole. Be that as it may, the differences in the average productivity of these countries' agriculture are significant enough to mitigate such issues insofar as one wishes to make broad inferences about the said differences.

		Finland	U.S.
Population in agriculture	1,000s	502	7,670
Share in agriculture ^(a)	percentage	73.5	44.1
Arable	hectares per capita	1.7	11.8
Harvest ^(b)	barrels per capita	15.5	84.6
Barley	barrels per hectare	8.5	13.5
Oats	barrels per hectare	8.4	15.1
Potato	barrels per hectare	7.3	29.7
Rye	barrels per hectare	11.5	6.5
Wheat	barrels per hectare	11.3	7.8

Table 6: Agricultural productivity comparison between Finland and the U.S., 1880. Source: Census Office, 1882, 1883; Statistical Office of Finland, 1882; U.S. Department of Agriculture, 2019; Vattula, 1983. Note: 1 bushel = 35.2 litres, 1 barrel = 4.2 bushels, 1 acre = 0.4 hectares. (a) Share of the occupied population; (b) all crops cultivated by acre or barrel (excluding flax and hemp in Finland).

Archives

Bank of Finland archives (Suomen Pankin arkisto)
National Archives (Kansallisarkisto)
Nordea history archives (Nordean historia-arkisto)

Others

His Imperial Majesty's Benignant Declaration on the consent tax that the Finnish Estates have taken upon themselves to pay in the years 1865, 1866 and 1867 (March 2, 1865/10)

His Imperial Majesty's Benignant Statute on certain guidelines when using the Benignant declaration on the consent tax that the Finnish estates have taken upon themselves to pay (April 20, 1865/23)

The Parliament's response N:o 48/1935 (AsK 1935 III)
The Parliament's response N:o 77/1943 (AsK 1943 I–II)
The Parliament's response N:o 16/1946 (AsK 1946 I–II)
The Parliament's response N:o 59/1947 (AsK 1947 I–II)
The Parliament's response N:o 4/1948 (AsK 1948 I–III)
The Parliament's response N:o 87/1950 (AsK 1950 III)
The Parliament's response N:o 18/1951 (AsK 1951 III)
The Parliament's response N:o 56/1952 (AsK 1952 III)
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